The Journal of Open Source Software

Den2Obj: A command-line tool for producing isosurfaces from electron density data files

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DOI: 10.21105/joss.10100

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Submitted: 11 May 2025
Published: 16 October 2026

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Summary

Electron density is a foundational quantity in electronic structure calculations. It describes the spatial distribution of electrons in molecules and materials, and is used to derive key properties such as total energy, electrostatic potential, and atomic forces. Because electron density is a three-dimensional scalar field, specialized visualization techniques, including contour plotting, isosurface generation, and volumetric rendering, are needed to interpret bonding, reactivity, and electron redistribution.

`Den2Obj` is a C++ command-line program for generating isosurfaces from electron density scalar fields. It supports input formats commonly used in electronic structure calculations, including CHGCAR and PARCHG files produced by VASP, as well as Gaussian Cube files. The resulting isosurfaces can be exported in widely supported geometry file formats, including STL (Stereolithography), PLY (Polygon File Format), and OBJ (Wavefront), enabling integration with visualization, rendering, and post-processing tools. In addition, `Den2Obj` can convert scalar fields to a documented native `.d2o` format, a lossless binary container that stores the grid and unit-cell metadata while selecting an efficient compression method automatically.

Statement of need

Isosurfaces play a central role in scientific research and engineering applications by providing an effective means of visualizing complex scalar fields and understanding intricate physical phenomena. An isosurface represents a set of points in a field where a scalar quantity, known as the *isovalue*, remains constant. Many software packages support isosurface generation, including `Open Data Explorer` (Thompson et al., 2004), `MATLAB` (The MathWorks Inc., 2022), `ParaView` (Ahrens et al., 2005), `VESTA` (Momma & Izumi, 2011), PyVista (Sullivan & Kaszynski, 2019), and chemistry-oriented tools built on such libraries, such as ChemVista (ChemParse, 2025). These packages and libraries are valuable for interactive inspection, scripting, and application-specific visualization workflows.

`Den2Obj` occupies a narrower niche: reproducible, non-interactive conversion of electronic-structure grid data into mesh and volume assets that can be used in automated analysis, documentation, and rendering pipelines. It directly reads formats commonly produced by `VASP` (Hafner, 2008), such as `CHGCAR` and `PARCHG`, as well as Gaussian (Frisch et al., 2016) Cube files. The resulting isosurfaces can be exported in widely used 3D geometry formats, including Stereolithography (`.stl`), Polygon File Format (`.ply`), and Wavefront (`.obj`).

Isosurface generation in `Den2Obj` is carried out using either the marching cubes algorithm (Lorensen & Cline, 1987) or the marching tetrahedra algorithm (Burke, 1994), both of which can be selected via command line arguments. These algorithms are implemented with OpenMP parallelization to leverage the performance of modern multi-core CPUs. When visualizing

wavefunctions rather than electron densities, it is often useful to generate separate isosurfaces for positive and negative lobes. `Den2Obj` accommodates this by enabling dual isosurface generation through a single command line argument.

An illustrative example is presented in Figure 1, which displays the canonical molecular orbitals of the benzene molecule. These orbitals were computed using the PyQInt program (Filot, 2025). Isosurfaces were generated via the marching cubes algorithm as implemented in Den2Obj, producing `.ply` files that were subsequently imported into Blender (*Blender - a 3D Modelling and Rendering Package*, 2018), along with the atomic coordinates of benzene, for rendering.

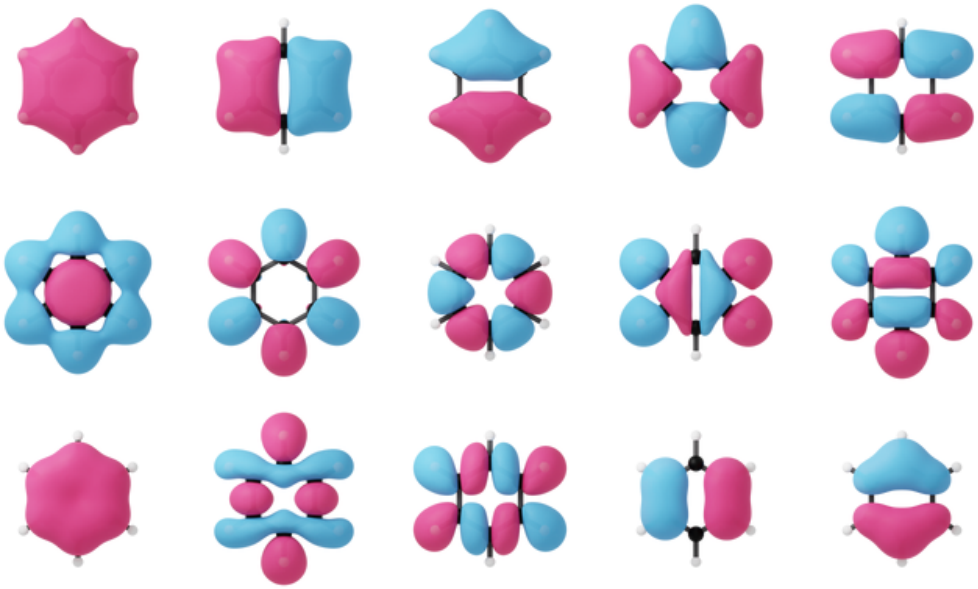


Figure 1: Isosurfaces of the first 15 canonical valence molecular orbitals of benzene.

For efficient research data management purposes, `Den2Obj` is also able to convert `CHGCAR` and `PARCHG` files to its own custom `.d2o` format. This format is one of the distinctive components of the software: it provides a documented, lossless binary representation of dense scalar fields, including unit-cell and grid metadata, so that large electronic-structure outputs can be archived and reloaded without reparsing text-based `CHGCAR` or `PARCHG` files. Upon conversion, the program explores multiple compression algorithms, namely `gzip`, `lzma`, `bzip2`, Zstandard (`zstd`) and Blosc, and uses the one that yields optimal results. These options cover Lempel-Ziv and Huffman-style compression (Huffman, 1952; Ziv & Lempel, 1977), block-sorting compression based on the Burrows-Wheeler transform (Fenwick, 1996), the Zstandard format (Collet & Kucherawy, 2021), and cache-aware block compression for numerical data (Alted, 2010). Such size reductions are expected when converting human-readable grid data to compressed binary storage; nevertheless, `.d2o` provides a convenient archival and intermediate representation, with files that can be as small as about 10% of the original `CHGCAR` or `PARCHG` file size.

Another distinguishing feature is direct conversion of electronic-structure scalar fields to OpenVDB (Museth, 2013) files for volumetric rendering in programs such as Blender. In contrast to the rendering of isosurfaces, the main advantage of volumetric rendering is that internal details and density variations are more prominently shown, providing a comprehensive and nuanced understanding of the scalar field. In a way, volumetric rendering lies in between isosurfaces and contour plots in terms of visualizing a scalar field. An example for the molecular orbitals of benzene is provided in Figure 2. For demonstration and testing purposes, also a scalar field generator functionality is included that can create a number of relevant scalar fields

to test the algorithms on.

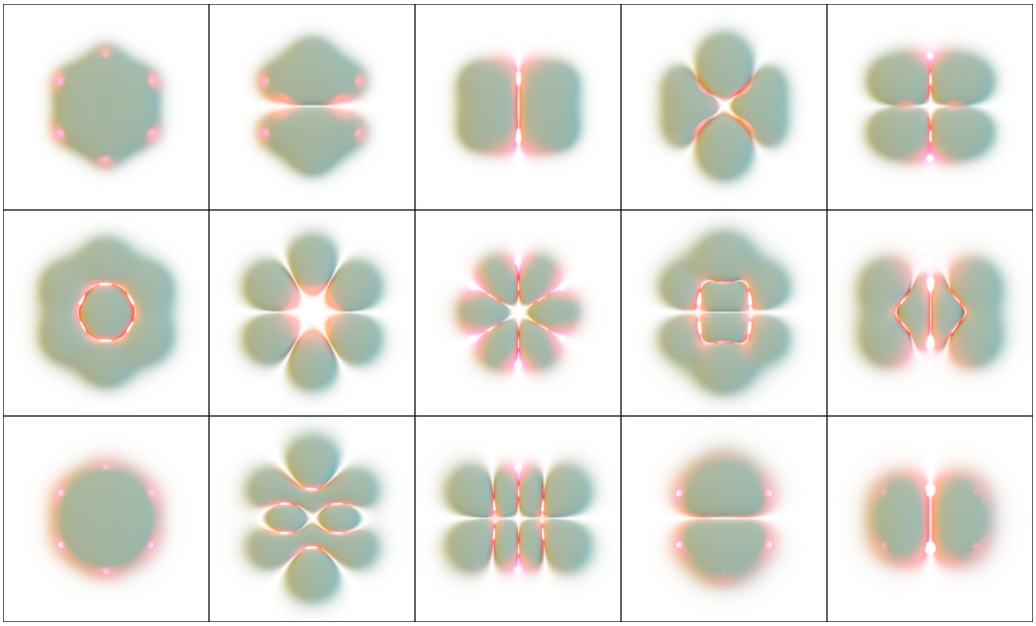


Figure 2: Volumetric rendering of the electron density associated with the first 15 canonical valence molecular orbitals of benzene using the OpenVDB format.

`Den2Obj` requires a relatively small set of dependencies: Eigen3 (Guennebaud et al., 2010), Boost (Koranne, 2011), `TCLAP` (Smoot et al., 2009), and compression libraries for `gzip`, `lzma`, `bzip2`, Zstandard and Blosc. Creation of VDB files requires the presence of the OpenVDB library (Museth, 2013). The user can select during compilation whether they want to include this functionality or not. `Den2Obj` is designed to be used by researchers and students working in computational materials modelling and quantum chemistry. The present package builds on an earlier command-line implementation of the same workflow. That earlier implementation was used in the studies reported in (Filot et al., 2016; Su et al., 2016, 2018).

An extensive user guide including examples, compilation instructions, tutorials (including a rendering tutorial in Blender) and documentation of the command-line arguments, is available at https://den2obj.imc-tue.nl/.

Acknowledgements

This work was supported by the Netherlands Center for Multiscale Catalytic Energy Conversion, and NWO Gravitation program funded by the Ministry of Education, Culture and Science of the government of the Netherlands. The Netherlands Organization for Scientific Research is acknowledged for providing access to computational resources.

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